

Chapter 18, Problem 64P

Problem

Repeat the previous problem for the FM broadcasting band (88–108 MHz), assuming that the carrier frequencies are spaced 200 kHz apart.

Step-by-step solution

Step 1 of 1

Space available for the FM channel, 88 MHz to 108 MHz,

i.e.

$$\text{Total space} = 108 \text{ MHz} - 88 \text{ MHz} = 20 \text{ MHz} = 20 \times 10^6 \text{ Hz}$$

If the channels are spaced 200 KHz ($= 200 \times 10^3 \text{ Hz}$) apart, then

$$\text{Total number of stations} = \frac{20 \times 10^6 \text{ Hz}}{200 \times 10^3 \text{ Hz}} = 100$$

or, 100 Stations